



BOARD QUESTION PAPER : MARCH 2022

MATHEMATICS AND STATISTICS

Time: 3 Hrs.

Max. Marks: 80

General instructions:*The question paper is divided into FOUR sections.*

- (1) **Section A:** Q.1 contains **Eight** multiple choice type of questions, each carrying **Two** marks.
Q.2 contains **Four** very short answer type questions, each carrying **one** mark.
- (2) **Section B:** Q.3 to Q. 14 contain **Twelve** short answer type questions, each carrying **Two** marks. (Attempt any **Eight**)
- (3) **Section C:** Q.15 to Q. 26 contain **Twelve** short answer type questions, each carrying **Three** marks. (Attempt any **Eight**)
- (4) **Section D:** Q.27 to Q. 34 contain **Eight** long answer type questions, each carrying **Four** marks. (Attempt any **Five**)
- (5) Use of log table is allowed. Use of calculator is not allowed.
- (6) Figures to the right indicate full marks.
- (7) Use of graph paper is not necessary. Only rough sketch of graph is expected.
- (8) For each multiple choice type of question, it is mandatory to write the correct answer along with its alphabet, e.g. (a)..... / (b)..... / (c)..... / (d)....., etc. No marks shall be given, if ONLY the correct answer or the alphabet of correct answer is written. Only the first attempt will be considered for evaluation.
- (9) Start answer to each section on a new page.

SECTION – A**Q.1. Select and write the correct answer for the following multiple choice type of questions: [16]**

- (i) The negation of $p \wedge (q \rightarrow r)$ is _____.
(a) $\sim p \wedge (\sim q \rightarrow \sim r)$ (b) $p \vee (\sim q \vee r)$
(c) $\sim p \wedge (\sim q \rightarrow r)$ (d) $p \rightarrow (q \wedge \sim r)$ (2)
- (ii) In $\triangle ABC$ if $c^2 + a^2 - b^2 = ac$, then $\angle B =$ _____.
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{6}$ (2)
- (iii) Equation of line passing through the points $(0, 0, 0)$ and $(2, 1, -3)$ is _____.
(a) $\frac{x}{2} = \frac{y}{1} = \frac{z}{-3}$ (b) $\frac{x}{2} = \frac{y}{-1} = \frac{z}{-3}$
(c) $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ (d) $\frac{x}{3} = \frac{y}{1} = \frac{z}{2}$ (2)
- (iv) The value of $\hat{i} \cdot (\hat{j} \times \hat{k}) + \hat{j} \cdot (\hat{k} \times \hat{i}) + \hat{k} \cdot (\hat{i} \times \hat{j})$ is _____.
(a) 0 (b) -1 (c) 1 (d) 3 (2)
- (v) If $f(x) = x^5 + 2x - 3$, then $(f^{-1})'(-3) =$ _____.
(a) 0 (b) -3 (c) $-\frac{1}{3}$ (d) $\frac{1}{2}$ (2)
- (vi) The maximum value of the function $f(x) = \frac{\log x}{x}$ is _____.
(a) e (b) $\frac{1}{e}$ (c) e^2 (d) $\frac{1}{e^2}$ (2)
- (vii) If $\int \frac{dx}{4x^2 - 1} = A \log \left(\frac{2x-1}{2x+1} \right) + c$, then $A =$ _____.
(a) 1 (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$ (2)





(viii) If the p.m.f of a r.v. X is

$$P(x) = \frac{c}{x^3}, \text{ for } x = 1, 2, 3$$

= 0, otherwise,

then $E(X) =$ _____

(a) $\frac{216}{251}$

(b) $\frac{294}{251}$

(c) $\frac{297}{294}$

(d) $\frac{294}{297}$

(2)

Q.2. Answer the following questions:

[4]

(i) Find the principal value of $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

(1)

(ii) Write the separate equations of lines represented by the equation $5x^2 - 9y^2 = 0$

(1)

(iii) If $f'(x) = x^{-1}$, then find $f(x)$

(1)

(iv) Write the degree of the differential equation

$$(y''')^2 + 3(y'') + 3xy' + 5y = 0$$

(1)

SECTION - B

Attempt any EIGHT of the following questions:

[16]

Q.3. Using truth table verify that:

$$(p \wedge q) \vee \sim q \equiv p \vee \sim q$$

(2)

Q.4. Find the cofactors of the elements of the matrix $\begin{bmatrix} -1 & 2 \\ -3 & 4 \end{bmatrix}$

(2)

Q.5. Find the principal solutions of $\cot \theta = 0$

(2)

Q.6. Find the value of k , if $2x + y = 0$ is one of the lines represented by $3x^2 + kxy + 2y^2 = 0$

(2)

Q.7. Find the cartesian equation of the plane passing through $A(1, 2, 3)$ and the direction ratios of whose normal are 3, 2, 5.

(2)

Q.8. Find the cartesian co-ordinates of the point whose polar co-ordinates are $\left(\frac{1}{2}, \frac{\pi}{3}\right)$.

(2)

Q.9. Find the equation of tangent to the curve $y = 2x^3 - x^2 + 2$ at $\left(\frac{1}{2}, 2\right)$.

(2)

Q.10. Evaluate: $\int_0^{\frac{\pi}{4}} \sec^4 x \, dx$

(2)

Q.11. Solve the differential equation $y \frac{dy}{dx} + x = 0$

(2)

Q.12. Show that function $f(x) = \tan x$ is increasing in $\left(0, \frac{\pi}{2}\right)$.

(2)

Q.13. From the differential equation of all lines which makes intercept 3 on x -axis.

(2)

Q.14. If $X \sim B(n, p)$ and $E(X) = 6$ and $\text{Var}(X) = 4.2$, then find n and p .

(2)

SECTION - C

Attempt any EIGHT of the following questions:

[24]

Q.15. If $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$, then find the value of x .

(3)

Q.16. If angle between the lines represented by $ax^2 + 2hxy + by^2 = 0$ is equal to the angle between the lines represented by $2x^2 - 5xy + 3y^2 = 0$, then show that $100(h^2 - ab) = (a + b)^2$.

(3)

Q.17. Find the distance between the parallel lines $\frac{x}{2} = \frac{y}{-1} = \frac{z}{2}$ and $\frac{x-1}{2} = \frac{y-1}{-1} = \frac{z-1}{2}$.

(3)





- Q.18.** If A (5, 1, p), B(1, q, p) and C(1, -2, 3) are vertices of a triangle and $G\left(r, \frac{-4}{3}, \frac{1}{3}\right)$ is its centroid, then find the values of p, q, r by vector method. (3)
- Q.19.** If $A(\vec{a})$ and $B(\vec{b})$ be any two points in the space and $R(\vec{r})$ be a point on the line segment AB dividing it internally in the ratio m : n then prove that $\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$. (3)
- Q.20.** Find the vector equation of the plane passing through the point A(-1, 2, -5) and parallel to the vectors $4\hat{i} - \hat{j} + 3\hat{k}$ and $\hat{i} + \hat{j} - \hat{k}$. (3)
- Q.21.** If $y = e^{\tan^{-1}x}$, then show that $(1+x^2)\frac{d^2y}{dx^2} + (2x-m)\frac{dy}{dx} = 0$ (3)
- Q.22.** Evaluate: $\int \frac{dx}{2+\cos x - \sin x}$ (3)
- Q.23.** Solve $x + y \frac{dy}{dx} = \sec(x^2 + y^2)$ (3)
- Q.24.** A wire of length 36 meters is bent to form a rectangle. Find its dimensions if the area of the rectangle is maximum. (3)
- Q.25.** Two dice are thrown simultaneously. If X denotes the number of sixes, find the expectation of X. (3)
- Q.26.** If a fair coin is tossed 10 times. Find the probability of getting at most six heads. (3)

SECTION - D**Attempt any FIVE of the following questions:****[20]**

- Q.27.** Without using truth table prove that $(p \wedge q) \vee (\sim p \wedge q) \vee (p \wedge \sim q) \equiv p \vee q$ (4)
- Q.28.** Solve the following system of equations by the method of inversion $x - y + z = 4, 2x + y - 3z = 0, x + y + z = 2$ (4)
- Q.29.** Using vectors prove that the altitudes of a triangle are concurrent. (4)
- Q.30.** Solve the L.P.P. by graphical method,
Minimize $z = 8x + 10y$
Subject to $2x + y \geq 7,$
 $2x + 3y \geq 15,$
 $y \geq 2, x \geq 0$ (4)
- Q.31.** If $x = f(t)$ and $y = g(t)$ are differentiable functions of t so that y is differentiable function of x and $\frac{dx}{dt} \neq 0$, then prove that:
 $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$
Hence find $\frac{dy}{dx}$ if $x = \sin t$ and $y = \cos t$. (4)
- Q.32.** If u and v are differentiable function of x, then prove that:
 $\int uv \, dx = u \int v \, dx - \int \left[\frac{du}{dx} \int v \, dx \right] dx$
Hence evaluate $\int \log x \, dx$ (4)
- Q.33.** Find the area of region between parabolas $y^2 = 4ax$ and $x^2 = 4ay$. (4)
- Q.34.** Show that: $\int_0^{\frac{\pi}{4}} \log(1 + \tan x) \, dx = \frac{\pi}{8} \log 2$ (4)

